Cambridge (CIE) IGCSE Maths
Paper 4: Calculator (Set A)
Extended

Friday 25 April 2025

Afternoon (Time: 2 hours 0 minutes)

Total marks

/ 100

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You will need geometrical instruments.
- You should use a scientific calculator where appropriate.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π, use either your calculator value or 3.142.

Materials

- List of formulas

Scan here to mark your mock exam
or visit the mock exams landing page for this course

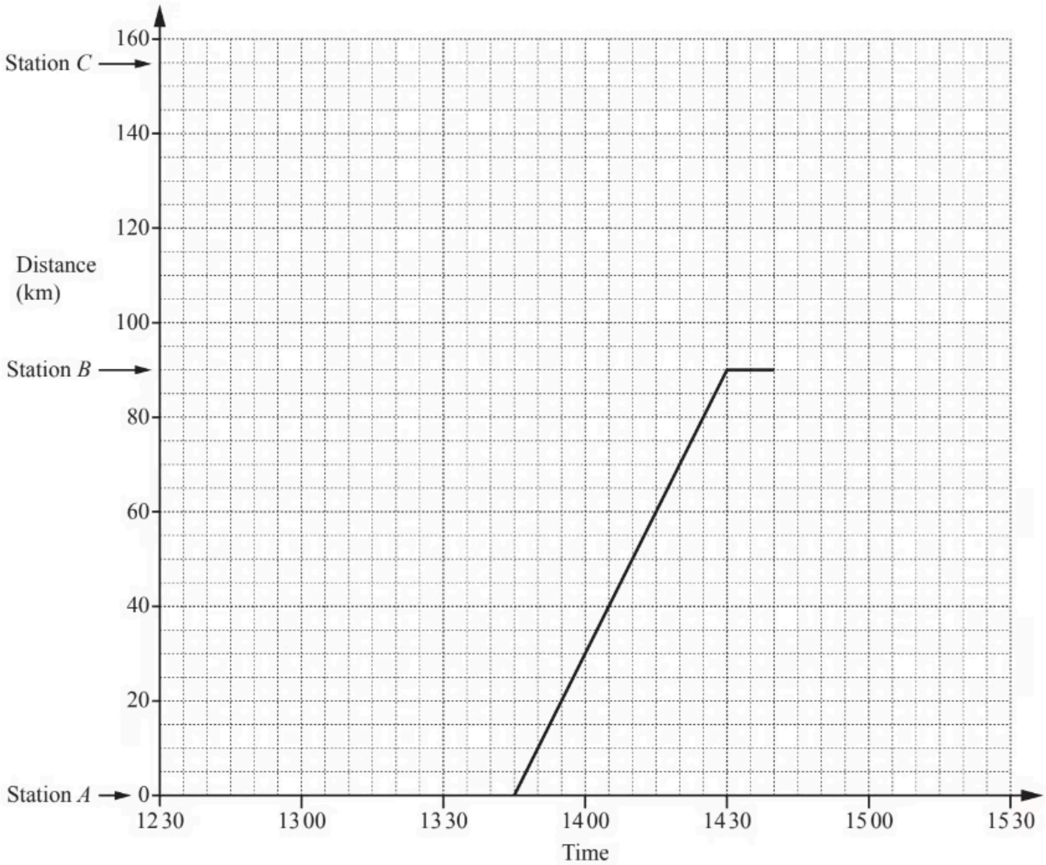
1 **Using a ruler and pair of compasses only**, construct a triangle with sides 5cm, 8cm and 10 cm.
Leave in your construction arcs.

(2 marks)

2 Simplify $(4xy^2)^3$.

(2 marks)

3 (a) The travel graph shows part of a train journey between station A and station C.



i) Calculate, in km/h, the speed of the train between station A and station B.

......................km/h [2]

ii) The train leaves station B at 14 40.

For how many minutes did the train stop at station B?

.........................min [1]

iii) The train travels at a constant speed between station B and station C, arriving at 15 20.

Complete the travel graph for the journey between station B and station C.

[1]

iv) On which part of the journey was the train travelling faster?

Between station .... and station .... [1]

(5 marks)

(b) Another train leaves station C at 12 45. It travels to station A at a constant speed of 62km/h without stopping at station B.

i) Work out how long, in hours and minutes, this journey takes.

.... h ....min [2]

ii) Write down the time this train arrives at station A.

[1]

iii) On the grid, show the journey of this train.

[1]

iv) Find the distance from station A when the two trains pass each other.

.... km [1]

(5 marks)

4 (a) Gabriela designs the seating layout for a new theatre.
There are three sections of seats, A, B and C.

Section A has 152 seats.
Section B has 171 seats.
Section C has 57 seats.

Write down and simplify the ratio of the number of seats in each section A : B : C.

A : B : C = ... : .... : ....

(2 marks)

(b) For a concert in the theatre, the ticket prices are in the ratio

A : B : C = 9 : 7 : 4.

A ticket for Section C costs $6.

i) Show that a ticket for Section B costs $10.50 .

[1]

ii) Find the cost of a ticket for Section A.

$ .... [1]

iii) The table shows the number of tickets sold in each section.

Section	Number of tickets sold
A	120
B	136
C	30

Calculate the total amount received from the ticket sales.

$ .... [3]

iv) The concert costs $4500 to organise.

Calculate the amount received from the ticket sales as a percentage of the $4500.

[1]

(6 marks)

5 Gino invests $6000 for 5 years at a rate of 1.2% per year compound interest.

Calculate the value of his investment at the end of the 5 years. Give your answer correct to the nearest dollar.

$ ....

(3 marks)

6 At a bus stop

- a red bus arrives every 18 minutes

and

- a blue bus arrives every 24 minutes.

At 1047 a red bus and a blue bus arrive.

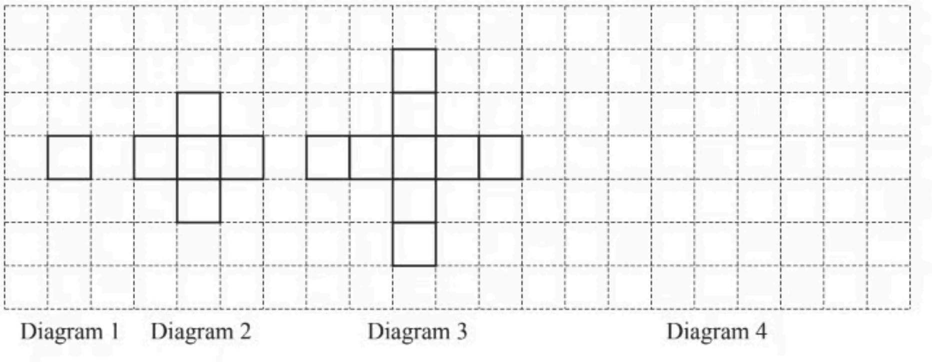
Find the next time when a red bus and a blue bus arrive together.

(3 marks)

7 The grid shows the first three diagrams in a sequence.

Each diagram is made using identical small squares.

Each square has sides that are 1 unit long.



i) On the grid, draw Diagram 4.

[1]

ii) Complete the table.

Diagram number	1	2	3	4
Perimeter	4	12	20	

[1]

iii) Find an expression, in terms of n, for the perimeter of Diagram n.

[2]

iv) For one of the diagrams in the sequence the perimeter is 300 units.

Work out its Diagram number.

[2]

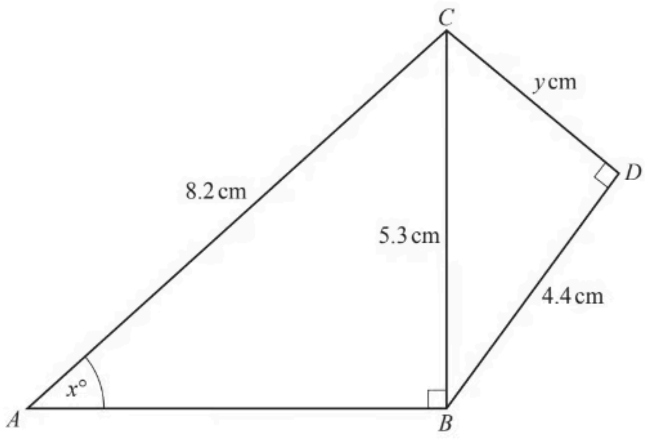
v) **Diagram 3** is drawn on a piece of card.
The side of each small square is 7cm.
The diagram is the net of an open box.

Calculate the volume of this box.
Give the units of your answer.

[3]

(9 marks)

8 (a)



NOT TO SCALE

Triangles ABC and BCD are both right-angled triangles.

Calculate the value of y.

y = ...

(3 marks)

(b) Calculate the value of x.

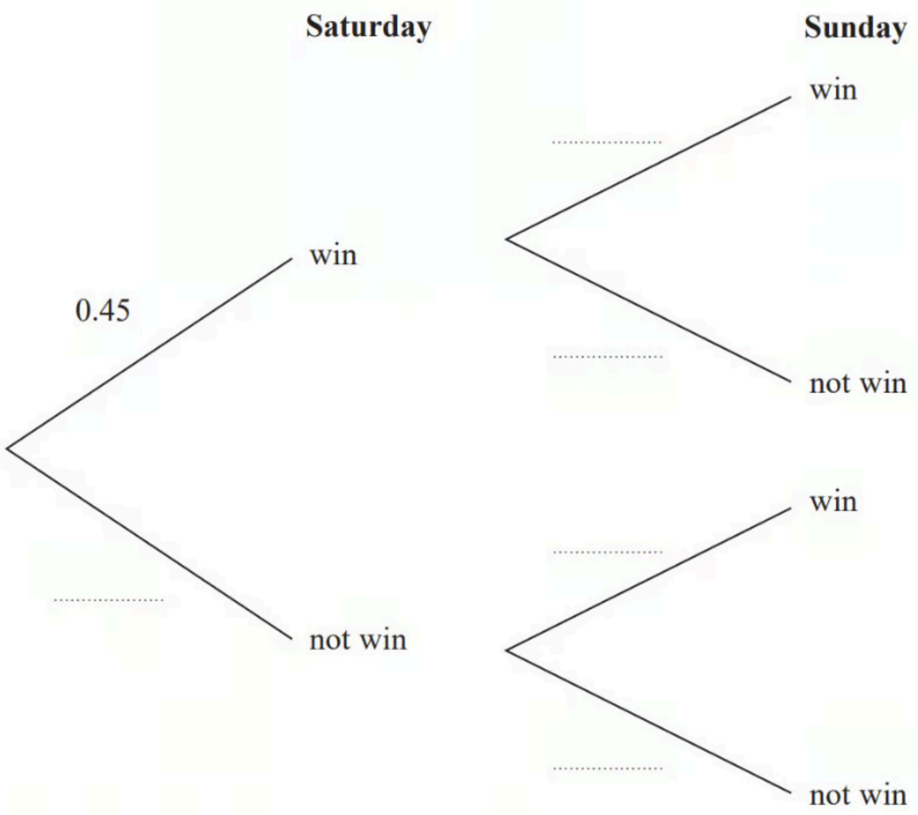
x = ...

(2 marks)

9 (a) A darts team is going to play a match on Saturday and on Sunday. The probability that the team will win on Saturday is 0.45

If they win on Saturday, the probability that they will win on Sunday is 0.67 If they do **not** win on Saturday, the probability that they will win on Sunday is 0.35

Complete the probability tree diagram.

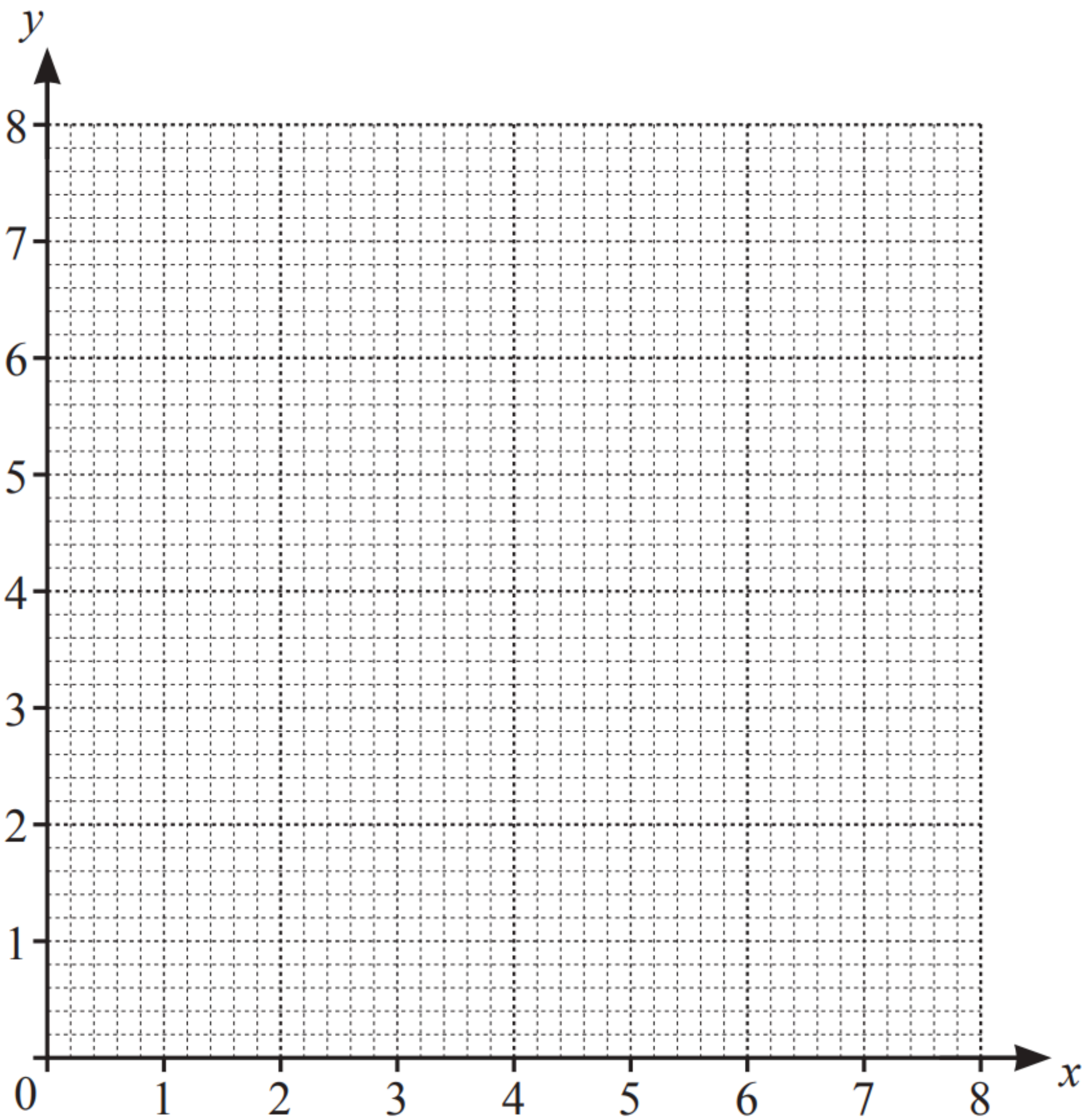


(2 marks)

(b) Find the probability that the team will win exactly one of the two matches.

(3 marks)

10



By drawing suitable lines and shading unwanted regions, find the region, R, where

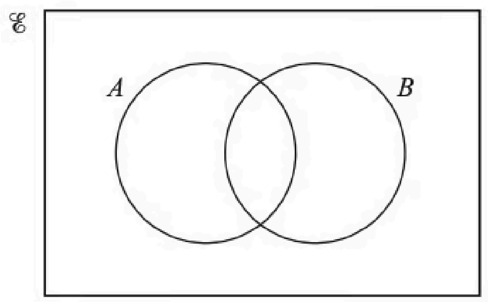
$$x \geq 2, \quad y \geq x \quad \text{and} \quad 2x + y \leq 8.$$

(5 marks)

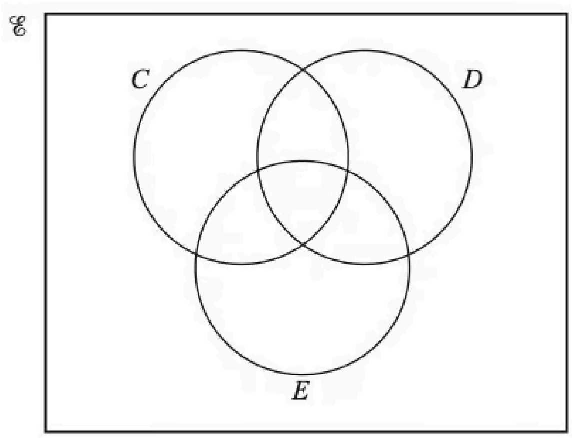
11 Factorise $3x + 8y - 6ax - 16ay$.

(2 marks)

12 In each Venn diagram, shade the required region.



$(A \cup B)'$



$(C \cap D') \cup E$

(2 marks)

13 The table shows information about the times, t seconds, taken by each of 100 students to solve a puzzle.

Time (t s)	$0 < t \leq 10$	$10 < t \leq 15$	$15 < t \leq 20$	$20 < t \leq 40$	$40 < t \leq 75$
Frequency	9	18	22	30	21

Calculate an estimate of the mean time.

.... s

(4 marks)

14 Make x the subject of the formula.

$$x = \frac{3 + x}{y}$$

$x =$ ....

(3 marks)

15 The coordinates of *P* are $(-3, 8)$ and the coordinates of *Q* are $(9, -2)$.

Calculate the length *PQ*.

(3 marks)

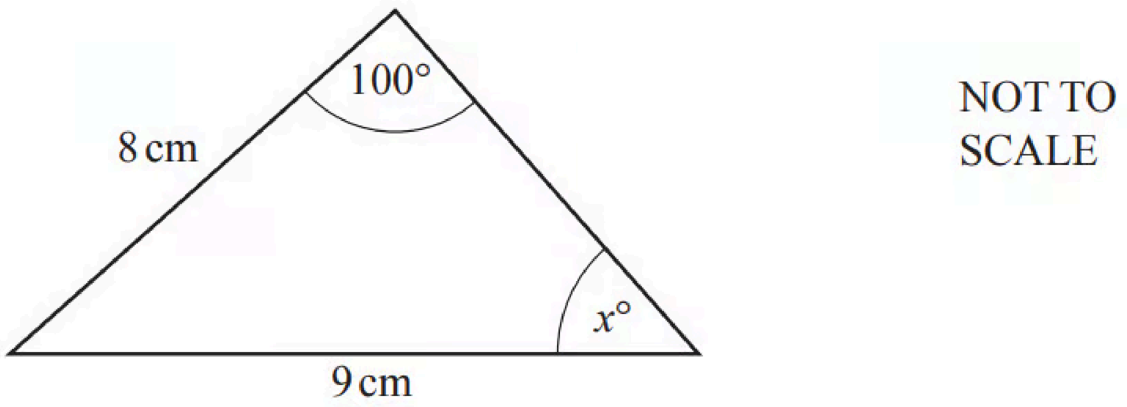
16 *A* is the point $(2, 3)$ and *B* is the point $(7, -5)$.

Find the equation of the line through *A* that is perpendicular to *AB*.
Give your answer in the form $y = mx + c$.

$y =$ ....

(4 marks)

17 (a)



Calculate the value of x.

x = ...

(3 marks)

(b) Calculate the area of the triangle.

.... cm^2

(3 marks)

18 i) Calculate the external curved surface area of a cylinder with radius 8 m and height 19 m.

.... m^2 [2]

ii) This surface is painted at a cost of $0.85 per square metre.

Calculate the cost of painting this surface.

$ .... [2]

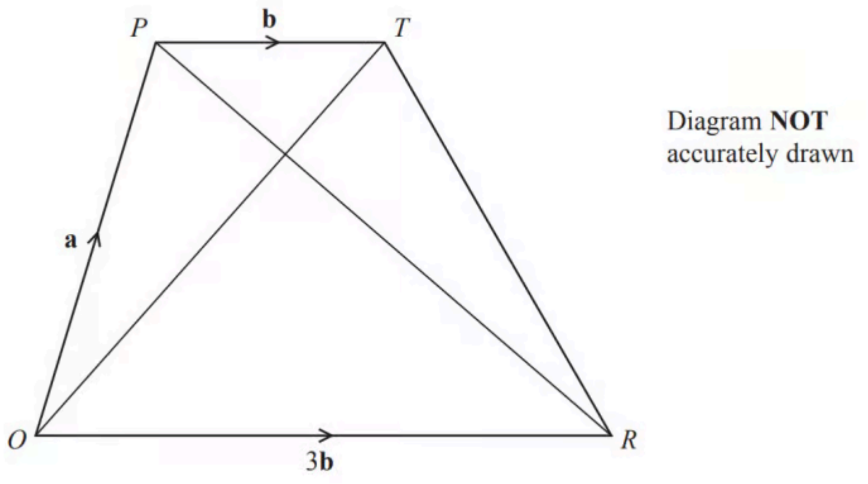
(4 marks)

19 A curve has equation $y = x^3 + \frac{7}{2}x^2 - 2x + 9$

Find $\frac{\mathrm{d}y}{\mathrm{d}x}$.

(2 marks)

20 (a)



$OPTR$ is a trapezium.

$\overrightarrow{OP} = \mathbf{a}$
$\overrightarrow{PT} = \mathbf{b}$
$\overrightarrow{OR} = 3\mathbf{b}$

i) Find $\overrightarrow{OT}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

[1]

ii) Find $\overrightarrow{PR}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ Give your answer in its simplest form.

[1]

(2 marks)

(b) S is the point on PR such that $PS : SR$ = 1 : 3

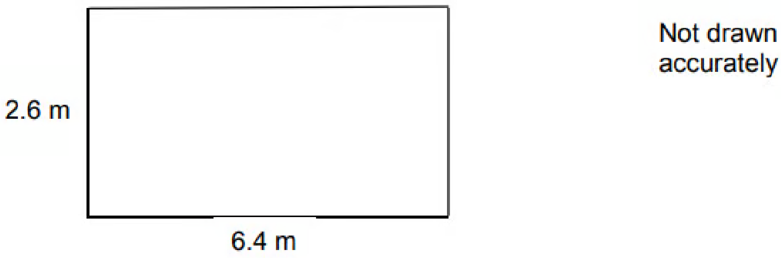
Find $\overrightarrow{OS}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.
Give your answer in its simplest form.

(2 marks)

(c) What does your answer to part (b) tell you about the position of point S?

(2 marks)

21 The dimensions of a rectangular floor are to the nearest 0.1 metres.



A force of 345 Newtons is applied to the floor.
The force is to the nearest 5 Newtons.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the upper bound of the pressure.
Give your answer to 4 significant figures.
You **must** show your working.

.......................N/m^2

(5 marks)

22 Simplify $\dfrac{x^2 - 25}{x^2 - 17x + 60}$.

(4 marks)

23 The nth term of a sequence is $4n^2 + n + 3$.
Find the value of n when the nth term is 498.

$n =$ ....

(3 marks)